BIOCHEMISTRY CUMULATIVE EXAMINATION

October 27, 1990

(25) 1. Draw a cartoon of an antibody molecule and label as many domains on the molecule as you can. What is the function of these labelled regions?

(25) 2. When antibodies and protein antigens are mixed, a precipitate usually forms. If the concentration of antigen is kept constant and the concentration of antibody increased, a graph of the precipitate vs. antibody is a bell-shaped curve. How do you account for the shape of this curve?

(25) 3. When antibodies are digested with papain, the proteolytic fragments no longer produce a precipitate when antigen is added. However, the antibody fragments can inhibit the precipitation of antigen with intact antibody. When antibodies are digested with pepsin, the fragments are still capable of precipitating with antigen. Assuming that there is no intact antibody left in either of the protease digests, explain these observations.

(25) 4. A normal population of antibodies has a wide range of binding affinities for a simple molecule like 2,4-dinitrophenol. Explain why this observation is consistent with the clonal expansion theory of antibody selection.